



Attachment of silver nanoparticles (AgNPs) onto thin-film composite (TFC) membranes through covalent bonding to reduce membrane biofouling



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ABSTRACT

Membrane biofouling has a negative impact on the membrane treatment performance. Silver nanoparticles (AgNPs) are well-known antimicrobial agent. Herein, AgNPs with approximately 15 nm in diameter were effectively attached to the surface of polyamide (PA) thin-film composite (TFC) membrane via covalent bonding, with cysteamine as a bridging agent. Scanning electron microscopy (SEM), energy dispersive X-ray spectroscopy (EDS) and cross-sectional transmission electron microscopy (TEM) studies all showed the immobilization of AgNPs. Compared with the pristine TFC membrane, thiol-terminated membrane (TFC-SH) and AgNPs grafted membrane (TFC-S-AgNPs) both showed a higher water flux with slightly lower salt rejection. At a constant transmembrane pressure of 300 psi, the water permeability of TFC-SH, TFC-S-AgNPs, and control TFC membranes was 70.6 ± 0.5 , 69.4 ± 0.3 , and 49.8 ± 1.7 L/m²h, respectively, while NaCl rejection was $93.4 \pm 0.1\%$, $93.6 \pm 0.2\%$, and $95.9 \pm 0.6\%$, respectively. TFC-S-AgNPs had an improved antibacterial ability to inhibit *E. coli* growth. The silver leaching from the TFC-S-AgNPs membrane surfaces was minimal, as tested by both batch and flow-through methods. The results successfully demonstrated that AgNPs could be grafted onto TFC via chemical bonding, leading towards the development of an advanced functional TFC membrane with anti-biofouling properties.

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1. Introduction

Membrane biofouling due to the microbial growth and biofilm formation is one of the most challenging issues in membrane separation processes [1]. Even when treated with disinfectants such as chlorine, some bacteria could still survive and migrate to the membrane surface and multiply [1,2]. Besides, the use of chlorine may result in formation of harmful disinfection byproducts and chlorine itself could accelerate degradation of polyamide TFC membranes [3,4]. In addition to biofouling that could decrease the membrane efficiency, the demand for pathogen-free clean water [5] is another driving force for the development of membranes with antimicrobial properties.

Silver ions and silver-based compounds have excellent biocidal properties and are widely used to prepare antimicrobial plastics, coatings, and wound and burn dressings, etc. [6,7]. To improve polymeric membrane's anti-biofouling property, many researchers introduced silver into membrane matrix. For example, Zdrov et al. [8] impregnated AgNPs into polysulfone (PSU) matrix to improve biofouling resistance and virus removal, and Chou et al.

[9] introduced AgNPs into cellulose acetate matrix for anti-bacterial applications. However, the AgNPs inside the bulk matrixes have weak resistance to washing and can be easily released in the form of Ag⁺ ions, so the membranes quickly lost their antimicrobial function [8–10].

Recently, efforts have been devoted to developing more effective methods to immobilize AgNPs into the membrane's barrier layer or graft them onto membrane surface through covalent bond formation. Lee et al. [3] and Kim et al. [11] immobilized AgNPs into the polyamide (PA) thin-film layer to improve antifouling properties of TFC membrane. Cao et al. [12] introduced AgNPs on the surface of sulfonated polyethersulfone membrane by using vitamin C as a reducing agent. Gunawan et al. [5] used polyethylene glycol-grafted multiwalled carbon nanotubes (MWNTs) as a connecting structure between AgNPs and hollow fiber membrane surface. In their study, AgNPs were firstly coated with MWNTs to prepare Ag/MWNTs, and then Ag/MWNTs were covalently bonded to the external surface of a chemically modified PAN hollow fiber membrane. In another study [13], AgNPs coated with positively charged polyethyleneimine (PEI) were immobilized in an oxygen plasma-modified PSU ultrafiltration (UF) membrane. These findings suggest that it is important to have the AgNPs located at the feed interface of membrane to allow a direct contact between silver and bacterial cells for best antimicrobial performance. To improve the durability of silver-containing membrane and

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simultaneously reduce the potential risks of releasing AgNPs and Ag^+ ions at high load to the environment [14,15], one approach is to chemically immobilize AgNPs on the membrane surface with slow silver release and long anti-biofouling characteristics.

In this study, the surface of freshly fabricated TFC membrane was first thiol-derivatized by reacting with $\text{NH}_2-(\text{CH}_2)_2-\text{SH}$ in ethanol solution, and then the freshly synthesized AgNPs were attached onto the membrane surface via the Ag–S chemical bonding. The membrane surface and cross-sectional characteristics were examined by scanning electron microscopy (SEM) and transmission electron microscopy (TEM), and surface chemistry was investigated using energy dispersive X-ray spectroscopy (EDS), attenuated total reflectance Fourier transform infrared (ATR FT-IR) spectroscopy, and Raman spectroscopy. We also assessed the membrane performance including water flux and salt rejection, Ag^+ ion release in both batch and flow-through tests, and antibacterial properties of the membranes by using Gram-negative bacteria, *E. Coli* as a model organism.

2. Materials and methods

2.1. Materials

All chemicals were of ACS reagent grades, obtained from Sigma-Aldrich (silver nitrate (AgNO_3), sodium borohydride (NaBH_4), polyvinylpyrrolidone (PVP40), polysulfone (MW=35,000), N, N-dimethylformamide (DMF, 99.8%), m-phenylenediamine (MPD, > 99%), trimesoyl chloride (TMC, > 98.5%), and cysteamine ($\text{H}_2\text{N}-(\text{CH}_2)_2-\text{SH}$, $\geq 98.0\%$). Deionized (DI) water, obtained from a Milli-Q ultrapure water purification system (Synergy 185, 18.2 M Ω ·cm, EMD Millipore Corp., Billerica, MA), was used for solution preparation and filtration study.

2.2. Synthesis and characterization of AgNPs

To synthesize AgNPs, a method modified from our previous work was applied here [16]. An aliquot (50 ml) of 14 mM sodium borohydride (NaBH_4) solution was mixed with 900 ml of 0.06 wt% PVP40 water solution, and then 50 ml of 14 mM silver nitrate (AgNO_3) solution was rapidly injected into the vigorously stirred solution (500 rpm) at room temperature (22 °C). TEM (JEOL 1400, JEOL Ltd., Peabody, MA) coupled with a submicron particle size analyzer (DelsaTM Nano HC, Beckman Colter, Inc., Brea, CA) was used to determine the morphology and size distribution of the

freshly synthesized AgNPs. TEM samples of AgNPs were prepared by dropping the AgNPs suspension onto carbon coated copper grid and drying at the room temperature.

2.3. Preparation of TFC and TFC-S-AgNPs membrane

The schematic diagram of membrane fabrication process was presented in Fig. 1. The PSU support layer was fabricated by the phase inversion method using a 15 wt% PSU-DMF casting solution. The casting solution was stirred at 50 °C for 6 h, and kept overnight for degassing. The clear solution was spread on a glass plate and casted by casting knife (EQ-Se-KTQ-150, MTI Corp., Richmond, CA) to approximately 100 μm of film thickness. Then, glass plate was immediately immersed into a water bath (25 °C). The precipitated PSU support membrane was washed and stored in DI water for at least 24 h before use. For TFC membrane fabrication, the prepared PSU support layer was attached to a small glass plate with tape around the edge, and then immersed in a 2.0 wt% MPD-water solution for 3 min. Excess solution on the surface was removed by a rubber roller. Next, the MPD saturated PSU support layer was soaked in a 0.15 wt% of TMC-hexane solution for 30 s, resulting in the formation of a polyamide (PA) thin-film layer. The TFC membranes were rinsed with pure hexane and then ethanol before the surface grafting.

To prevent hydrolysis of acyl chlorides, a newly fabricated TFC membrane sample (10 cm^2) was immediately immersed in a $\text{H}_2\text{N}-(\text{CH}_2)_2-\text{SH}$ ethanol solution (20 mM, 50 ml) for various time durations (3, 6, 24 h). And then the membrane sample (labeled as TFC-SH) was moved out from the ethanol solution, rinsed with pure ethanol and DI water, and incubated with the PA active layer in contact with an AgNPs suspension (0.1 mM, 50 ml) for 12 h. Finally, the membrane sample (labeled as TFC-S-AgNPs) was rinsed with DI water and stored in DI water at 4 °C before use.

2.4. Membrane characterization and performance assessment

SEM and EDS analysis of membrane surface was conducted using a piece of membrane dried at room temperature. To obtain the TEM cross-section, the membranes were embedded in Epon resin (Eponate 12, Ted Pella, Inc., Redding, CA) and cut by Reichert-Jung Ultracut E ultramicrotome (Reichert, Inc. Depew, NY). The images were taken under 80 kV by using JEOL 1400. Hydrophilicity of membrane surface was assessed according to the pure water contact angle, which was measured by the sessile drop method on

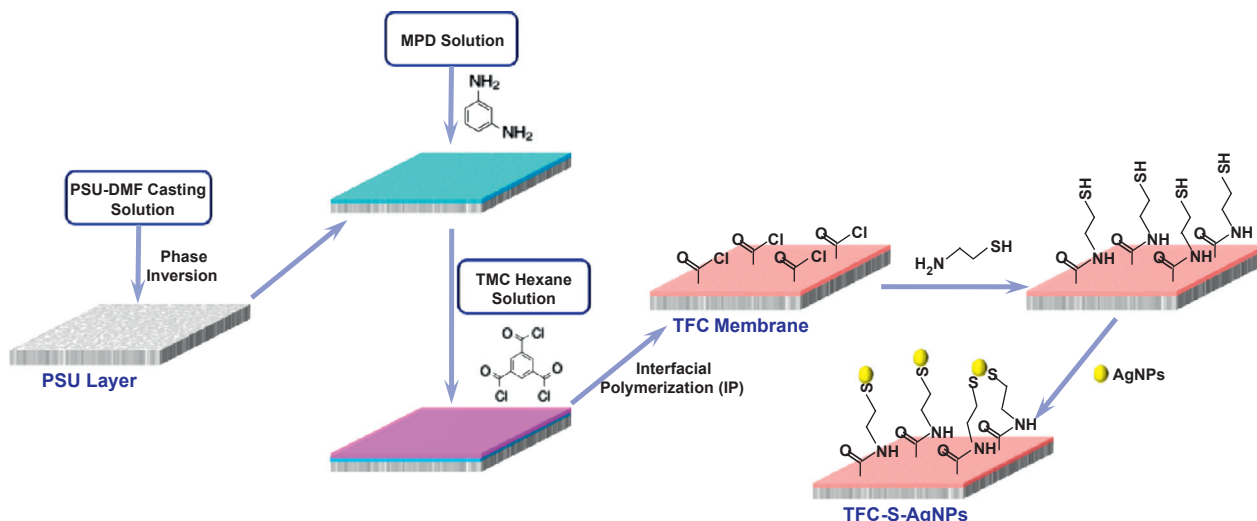


Fig. 1. Schematic diagram of immobilization of AgNPs onto the surface of PA TFC membrane.

a video contact angle system (VCA-2500 XE, AST products, Billerica, MA). At least six stabilized contact angles from different locations of each sample were obtained to calculate an averaged value of contact angles. The functional groups of membrane surface were identified by ATR FT-IR spectroscopy, which was conducted on the Nicolet 4700 FT-IR (Thermo Electron Corp., Waltham, MA) equipped with multi-reflection Smart Performer[®] ATR accessory. All spectra included the wave numbers from 500 to 4000 cm⁻¹ with 128 scans at a resolution of 1.0 cm⁻¹. An Invia Renishaw spectrometer system (RM1000 series, Gloucestershire, UK) equipped with a Leica DMLB microscope (Wetzlar, Germany) and 514.5 nm laser line was used to detect the thiol group on membrane surface.

A high pressure cross-flow filtration system was used to evaluate pure water flux and solute rejection, as described in our previous work [17]. Prior to test, each membrane (9.6 cm² of effective area) was compressed by DI water at 300 psi for 3 h. Pure water flux was measured by the weight of permeate water at a constant transmembrane pressure (TMP). The weight of the permeate water was recorded by a LabVIEW automated system (National Instruments LabVIEW 8.2 with Ohaus digital balance). After the flux test, NaCl salt solution of 2000 mg/L was added and the conductivity of feed and permeate solutions was measured by a conductivity/TDS meter (HACH Company, Loveland, CO). The test was conducted at 25 ± 1 °C, controlled by a water circulator (Isotemp 6200 R20F, Fisher Scientific, Inc., Pittsburgh, PA). The flux and rejection was calculated with Eqs. (1) and (2), respectively.

$$J = \frac{V_p}{At} \quad (1)$$

$$R = \left(1 - \frac{C_p}{C_f}\right) \times 100. \quad (2)$$

where J is the water flux (L/m²h), V_p the permeate volume (L), A the membrane area (m²), t the filtration time (h), R the rejection ratio, and C_p and C_f the conductivities of permeate and feed solution, respectively.

Releases of silver ions from the functionalized membranes were assessed via both batch and flow-through experiments. In the batch test, membrane samples were incubated in DI water (the ratio is 1 cm² membrane sample: 20 ml DI water) placed on an orbital shaker (4516, Forma Scientific, Inc., Marietta, OH) under 100 rpm and the water was replaced every 24 h. All water samples were collected and acidified by 1% HNO₃ and analyzed by an atomic absorption spectrophotometer (AA, model 210 VGP +220 GF, Buck Scientific, Inc., East Norwalk, CT). The silver release rate under the flow-through condition was assessed by driving DI water through the membrane at a constant TMP of 300 psi, and permeate was collected as a function of time and analyzed for the released silver concentration by AA.

2.5. Antibacterial assessment

E. coli MG1655 (ATCC 47076) was inoculated into liquid culture of Lysogeny broth (LB, 20 g/L) and incubated in an Isotemp incubator (Fisher Scientific, Inc., Pittsburgh, PA) with shaking at 150 rpm overnight under 37 °C. This culture was then centrifuged for 10 min at 2700 rpm. After decanting the supernatant, the pellet was resuspended in LB solution. The resulting bacterial culture was centrifuged again, followed by resuspension of the pellet in LB solution. The resulting cell suspension served as a bacterial stock solution, which was further diluted to approximately 10⁵ colony-forming units (CFU)/ml before use. Studies of the antibacterial properties of membrane samples, disk experiment, SEM investigation and biofilm growth test were carried out as detailed below.

In the disk experiment, aliquots (50 µL) of *E. coli* working solutions were spread, in triplicate, onto plates of nutrient agar. Membrane samples (diameter 1 cm and 3.7 cm) were then placed onto the nutrient agar plate with the active layer in contact with the agar surface. After incubation at 37 °C for 24 h, the colony forming cells under the membrane samples were examined.

During SEM investigation, membrane samples with size of 5 cm² were immersed in 25 mL of *E. coli* suspension (1.0 × 10⁵ CFU/mL) with shaking at 150 rpm for 5 h at room temperature. Then, the bacteria were fixed in 3 vol.% glutaraldehyde PBS solution for 5 h at 4 °C and rinsed with PBS solution several times to remove any remaining glutaraldehyde on the surface. After fixation, the samples were dehydrated with an ethanol series (25, 50, 75 and 100%), air-dried, and coated with platinum (20 mA, 30 s) using sputter coater for SEM observation.

Furthermore, a biofilm growth test was carried out to examine the anti-biofouling properties of membranes on a drip flow biofilm reactor (DFR110, Biosurface Technologies Co., Bozeman, MT). First, membrane samples (2.5 cm × 6.0 cm) were attached on a glass plate with active layer facing up and immersed into 25 mL of *E. coli* suspension (1.0 × 10⁵ CFU/mL) for 5 h. Then the membrane samples were moved from the suspension and gently rinsed with PBS solution for three times. The membranes were transferred into different test chambers and feed with LB solution (5 g/L) at 1.0 ml/min for 7 days. Finally, the growth of biofilm on membrane surface was examined.

3. Results and discussion

3.1. AgNPs characteristics

As shown by the TEM image in Fig. 2a, AgNPs were well dispersed and of spherical morphology. The average diameter was 14.8 nm based on the calculation by the software ImageJ from a total of 171 particles. The absorption spectra of the AgNPs suspension (Fig. 2b) showed a well-defined plasmon band at about 400 nm, characteristic of nanosized silver [18]. And the intensity of this peak increased proportionally with increasing AgNPs concentration. The particle size distribution was given in the upper right corner of Fig. 2b, and the average diameter of 15.3 ± 4.8 nm was consistent with that from the TEM analysis.

3.2. Membrane properties

The photos of original, thiol modified and silver immobilized TFC membranes are shown in Fig. 3a. After reaction with AgNPs suspension, the originally white-colored thiol-modified TFC membrane became yellowish, indicating the immobilization of AgNPs on the membrane surface. Based on the SEM images as shown in Fig. 3(b–e), all of these membrane surfaces had a “ridge and valley” structure characteristic of the PA thin-film layer [19,20]. There were no obvious differences among these membranes. Fig. 3f showed the SEM morphology of TFC-S–AgNPs membrane after it was used in filtration for 72 h under 300 psi. The “ridge and valley” structure was apparently flattened, which could be attributed to the compaction under the pressure.

The high resolution SEM images showed spherical particles located on the membrane surface, both before (Fig. 4a) and after (Fig. 4b) the filtration test. The particles ranged from 10 to 20 nm in diameter, about the same size with that of AgNPs. Furthermore, the EDS spectra of four spots located on membrane surface (Fig. 4) showed the presence of silver, confirming the attachment of AgNPs on the membrane surface. The weight percentage of silver at those four spots is 5.12%, 24.78%, 4.15% and 8.47%, respectively. The amount of AgNPs attached to the TFC-S–AgNPs surface was

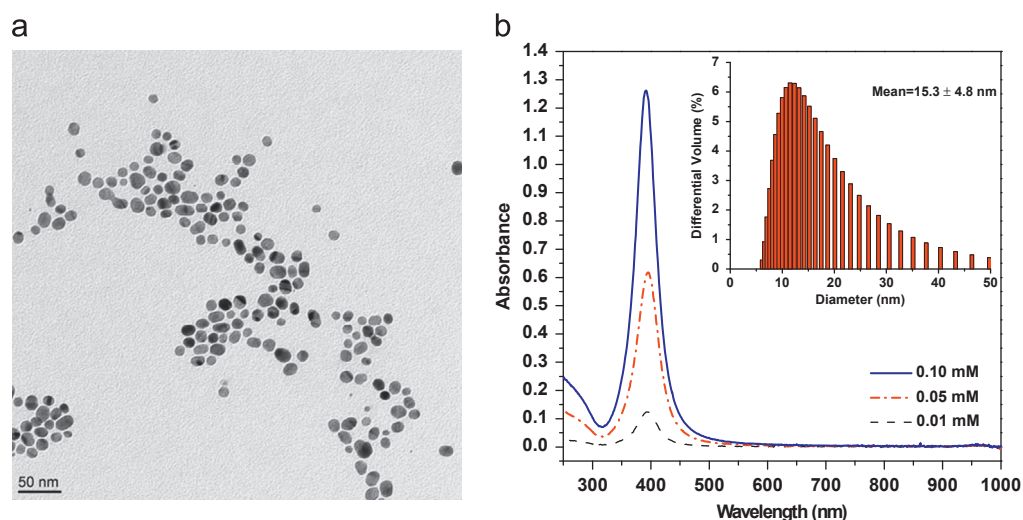


Fig. 2. TEM image of AgNPs (a) and UV-vis absorption spectrum and size distribution of AgNPs suspension (b).

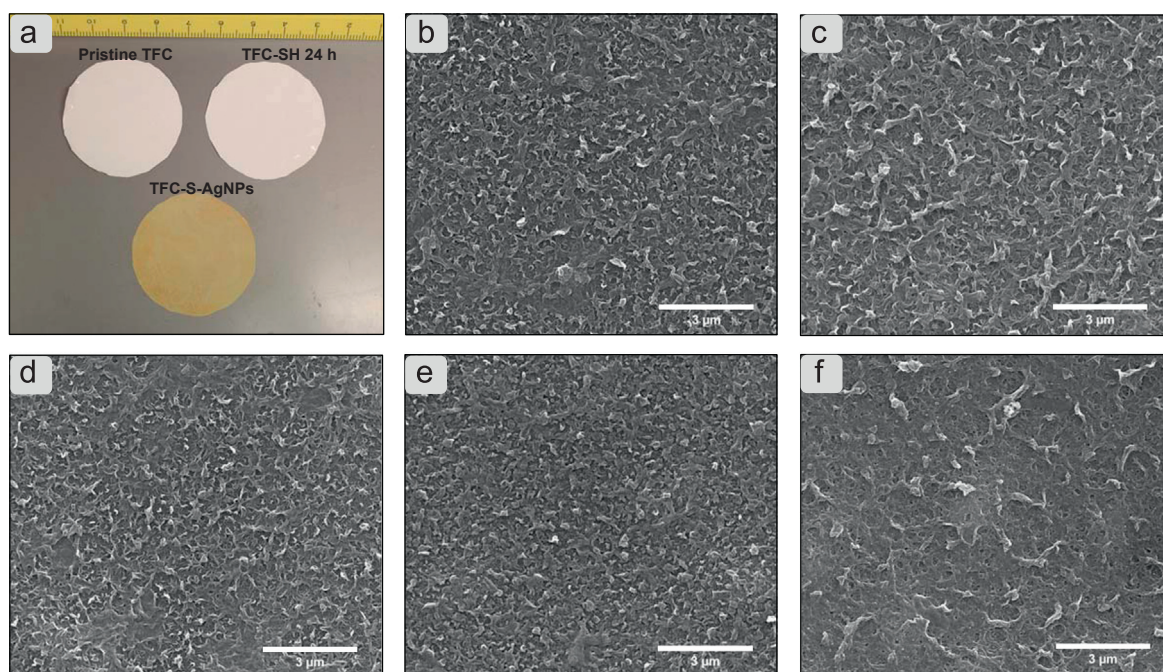


Fig. 3. Picture of membrane samples (a) and surface morphologies of pristine TFC (b), TFC-SH 6 h (c), TFC-SH 24 h (d), TFC-S-AgNPs (e) and TFC-S-AgNPs (after 72 h filtration) membrane (f). The scale bar in each SEM image is 3 μm.

further measured through acid leaching and AA analysis. The leaching test showed that there was 15.5 ± 2.6 μg of AgNPs that could be leached out on every square centimeter of membrane surface, corresponding to approximately 0.7% of the total mass of a membrane sample. Clearly this value is lower than the result from the EDS analysis. Considering the limited penetration depth of primary electrons in EDS (in the order of 2 μm for a carbon sample at an acceleration voltage of 20 kV [21]) and accumulation of AgNPs that occurred primarily on the membrane surface, it was reasonable to observe a higher amount of silver in EDS analysis than that by the leaching test.

Cross-sectional TEM images provided more detailed information about the distribution of AgNPs on membrane surfaces (Fig. 5). The images clearly delineated the 'ridge and valley' structure of the PA thin-film layer of a thickness between 100 and 200 nm, located on the porous PSU support layer.

A comparison between TFC and TFC-S-AgNPs indicated the presence of AgNPs on the surface of PA thin-film layer as dark spots, consistent with the SEM image and EDS data for the TFC-S-AgNPs membrane. It appeared that the amount of AgNPs remained the same on the TFC-S-AgNPs surface after 72 h of filtration test (Fig. 6e and f), suggesting a strong attachment of the AgNPs on the membrane surface. Similar to the SEM results, the TEM images showed a denser and smoother surface after the membrane was used for filtration due to compaction.

FT-IR spectra of various membrane surfaces are shown in Fig. 6. Peaks originated from the PA polymerization included 1660 cm^{-1} (amide I, C=O stretching vibrations of amide), 1547 cm^{-1} (amide II, in-plane N-H bending and C-N stretching vibrations), 1610 cm^{-1} (N-H stretching of amide), and 1450 cm^{-1} (C=O stretching and O-H bending of carboxylic acid) [22–24]. The peak at 768 cm^{-1} could be due to the C-Cl stretching [25]. After reaction

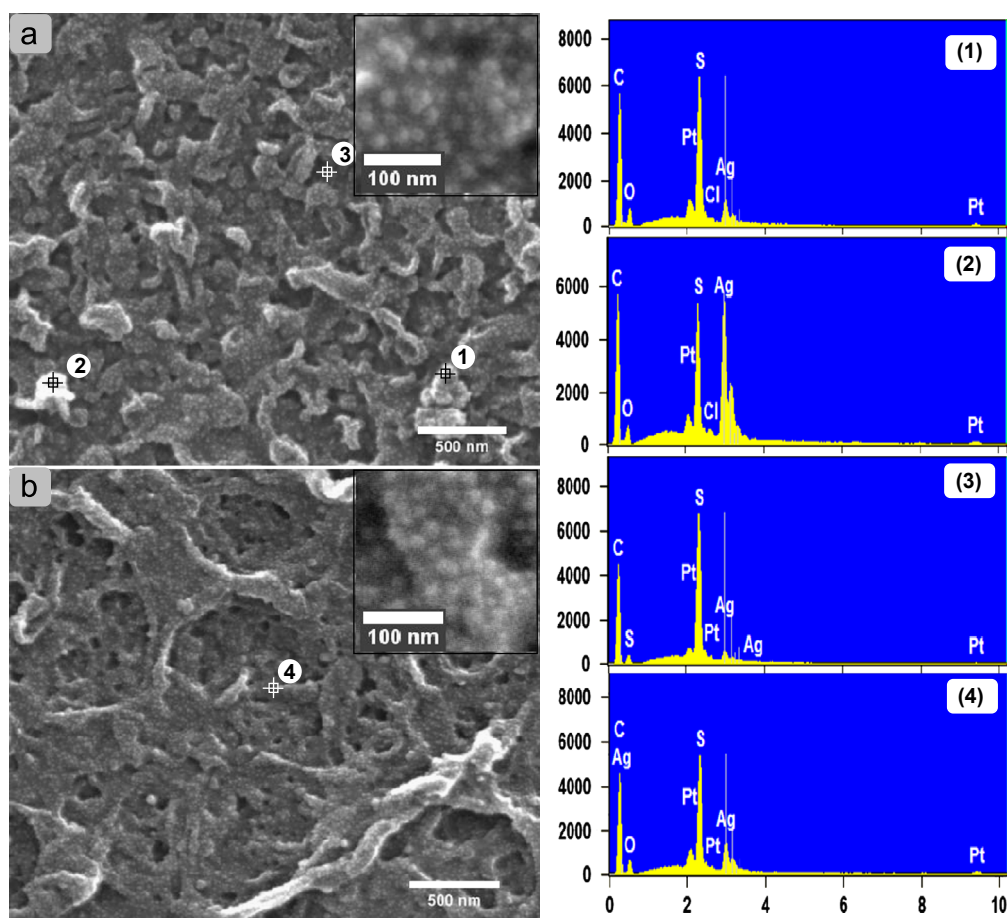


Fig. 4. High magnification SEM image and EDS spectra of TFC-S-AgNPs surface before filtration (a) and after 72 h filtration (b). Four spots were randomly chosen and labeled by numbers. The scale bar in each SEM image is 500 nm. Two higher magnification images were inserted in the upper right corner of the original images.

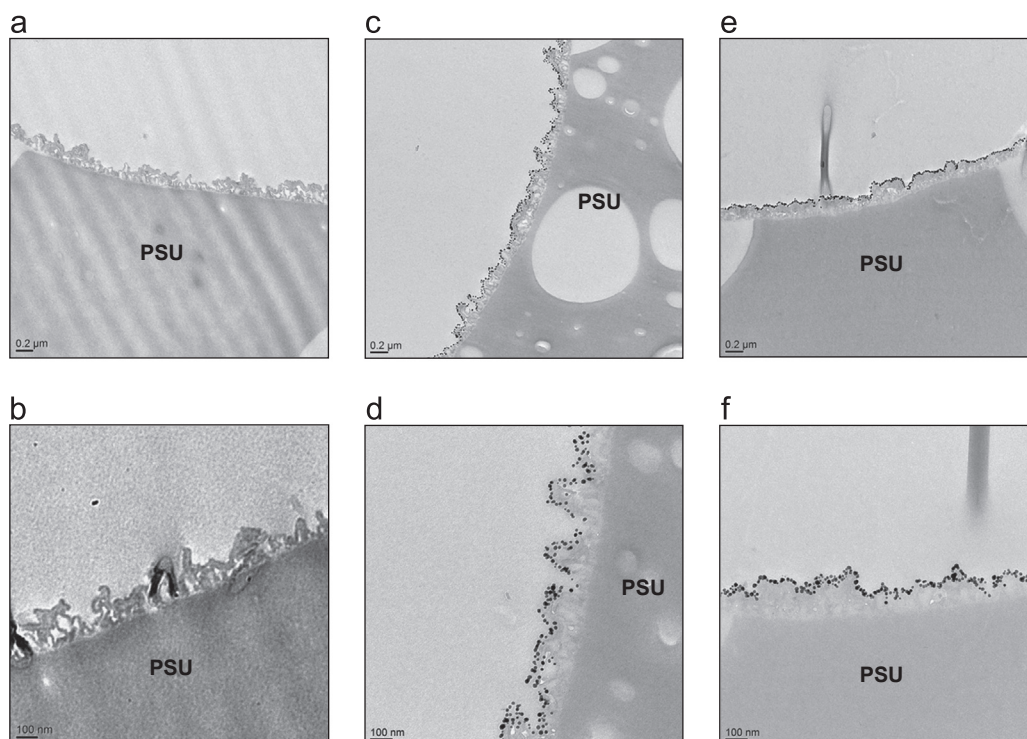


Fig. 5. TEM images of the cross-sections of (a,b) TFC membrane; (c,d) TFC-S-AgNPs membrane; (e,f) TFC-S-AgNPs membrane after 72 h filtration. Dark spots located on functionalized membrane surface are AgNPs.

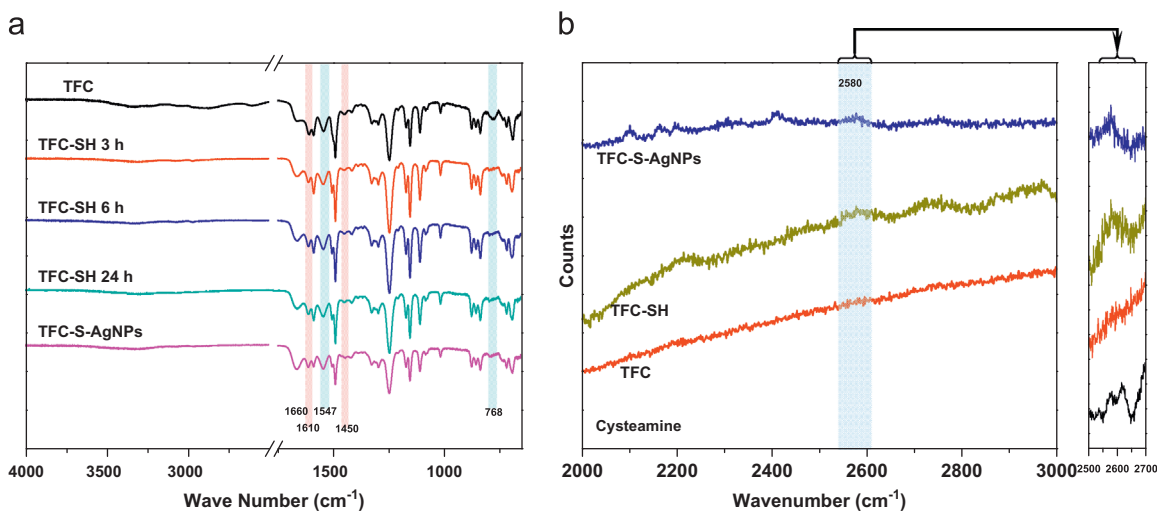


Fig. 6. Spectrum analysis of membrane surface: (a) ATR FT-IR; (b) Raman.

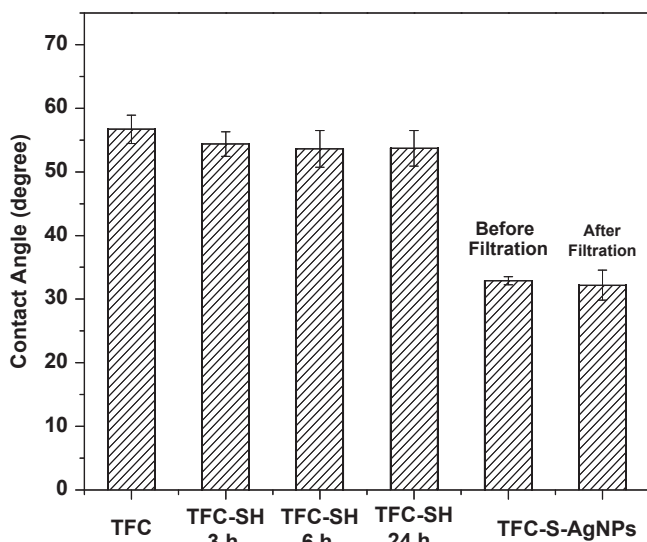


Fig. 7. Water contact angles of membrane surfaces.

between C–Cl and $\text{H}_2\text{N}-(\text{CH}_2)_2-\text{SH}$, the intensity of this peak decreased. However, the characteristic peak of S–H stretching located near $2540\text{--}2560\text{ cm}^{-1}$ [26–28] did not show up. This could be caused by two reasons: (i) the low sensitivity of thiol group in FT-IR due to the high polarizability of sulfur [27]; and (ii) the small amounts of thiol group on membrane surface.

Thiol group is highly polarizable and could produce a stronger Raman spectral activity than the infrared spectrum [29], so we explored to detect the thiol group on the membrane surface by Raman spectroscopy. As shown in Fig. 6b, both TFC-SH and TFC-S-AgNPs had a peak around 2580 cm^{-1} but not for the original TFC membrane, suggesting the existence of thiol group on the membrane surface [30].

In comparison with the original TFC membrane with a water contact angle of $56.7 \pm 2.2^\circ$, the contact angles of membrane with grafted $\text{NH}_2-(\text{CH}_2)_2-\text{SH}$ were essentially the same (Fig. 7), suggesting the existence of thiol groups on membrane surface did not significantly change the surface hydrophilicity. After grafting AgNPs, however, the contact angle decreased to $32.9 \pm 0.7^\circ$ (before filtration) and $32.2 \pm 2.4^\circ$ (after filtration). The decrease in contact angle by the grafted AgNPs, although it was somewhat more significant, is generally consistent with the

reports in the literature [8,11,31]. AgNPs were accumulated on the membrane surface in this study, which could have a more direct effect on the surface hydrophilicity.

3.3. Performance of membranes for water filtration

The pure water flux, salt rejection and permeate flux of membranes are presented in Fig. 8. At a constant TMP of 300 psi, the pure water flux increased from $67.1\text{ L/m}^2\text{h}$ (original TFC) to $90.6\text{ L/m}^2\text{h}$ (TFC-SH, 24 h) and $88.7\text{ L/m}^2\text{h}$ (TFC-S-AgNPs). When feeding with a 2000 mg/L NaCl solution, the permeate flux of the original TFC, TFC-SH (24 h), and TFC-S-AgNPs membranes were 49.8 ± 1.7 , 70.6 ± 0.5 , and $69.4 \pm 0.3\text{ L/m}^2\text{h}$, respectively, and NaCl rejection was $95.9 \pm 0.6\%$, $93.4 \pm 0.1\%$, and $93.6 \pm 0.2\%$, respectively.

The higher water flux and slightly lower salt rejection for the TFC-SH and TFC-S-AgNPs membranes could be due to the effects of ethanol solution used in the grafting process. Kulkarni et al. [32] also suggested that a mild solvent such as ethanol could increase the water flux of TFC membranes with no loss in rejection. Ethanol could result in membrane swelling and removal of small molecular fragments, so a more loose structure can be produced but still not enough to significantly decrease salt rejection.

3.4. Stability of AgNPs-containing membranes

The release rate of silver ions from the AgNPs grafted membrane would control the duration of the effectiveness of grafted AgNPs, so we examined the silver release in both batch and flow-through experiments. As presented in Fig. 9a, the initial silver ion release from the TFC-S-AgNPs membrane was around $0.37\text{ }\mu\text{g cm}^{-2}\text{ day}^{-1}$ and then declined steadily with time. After 7 days, the release rate leveled off to a level below $0.1\text{ }\mu\text{g cm}^{-2}\text{ day}^{-1}$. Based on the total amount of AgNPs on membrane surface (which was $15.5 \pm 2.6\text{ }\mu\text{g/cm}^2$), the amount leached out during 14 days of batch test accounted for only 12% of the total amount of silver on the membrane. Considering the relatively low release rate of around $0.1\text{ }\mu\text{g cm}^{-2}\text{ day}^{-1}$ at the steady state, the membrane is expected to be effective for months for its anti-bacterial properties due to the strong covalent bonding between the thiol groups and AgNPs.

In the flow-through experiment (Fig. 9b), silver ion concentration in the permeate side is very low, less than $5\text{ }\mu\text{g/L}$ in the initial 0.25 L of permeate and further decreased to less than $1\text{ }\mu\text{g/L}$ (detection limit) after about 2.0 L water permeating through the

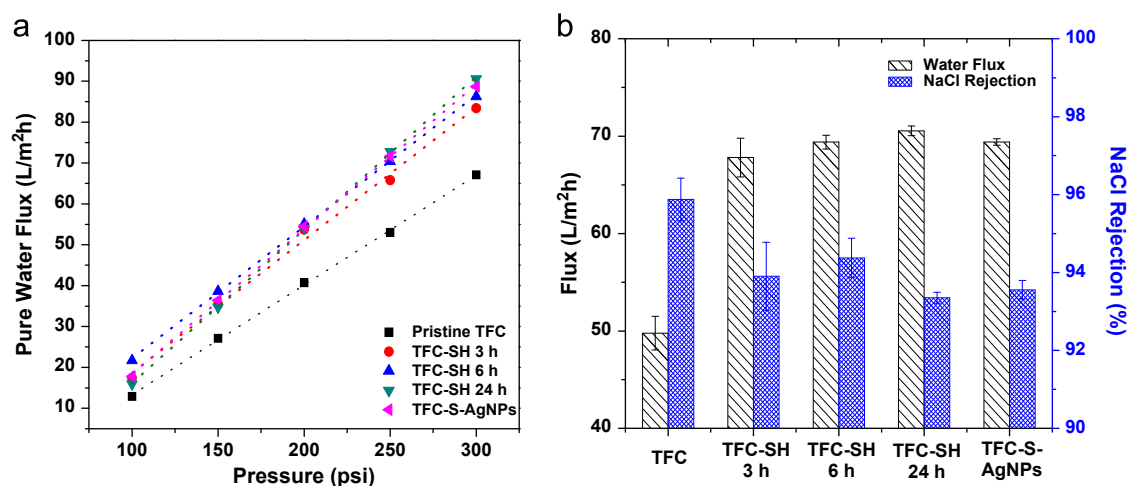


Fig. 8. Performance of membrane samples: (a) pure water flux; (b) NaCl rejection and solution flux (2000 mg/L NaCl solution was fed under 300 psi of TMP).

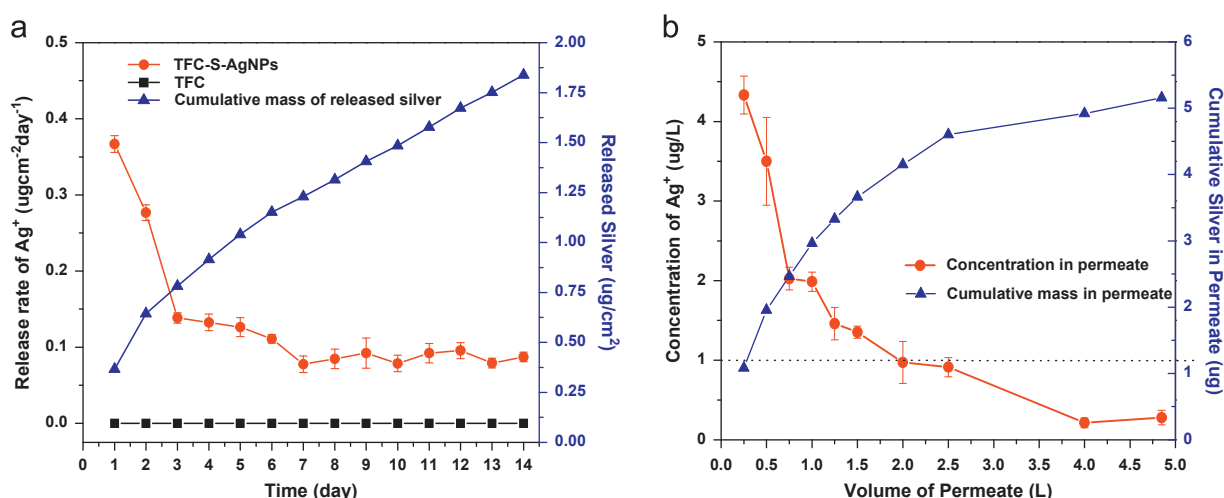


Fig. 9. Silver ion release from the batch (a) and flow-through (b) tests. During the batch test, membrane samples were incubated in 20 ml DI water under 100 rpm and the water was replaced every 24 h. During the flow-through test, DI water was driven through the membrane at a constant TMP of 300 psi.

membrane. Although the last several data points were already below the detection limit and may not be that accurate, we still gave out the calculated number based on the calibration curve to show the trend of concentration change. The National Secondary Drinking Water Regulations for silver is 100 $\mu\text{g/L}$, so the amount of silver released from the membrane, if there is any, is of no health concern. The cumulative amount of Ag⁺ ions in the 4.85 L of permeate (5.16 μg) accounted for only 3.5% of the total amount of silver in the membrane samples ($9.6 \text{ cm}^2 \times 15.5 \mu\text{g/cm}^2 = 148.8 \mu\text{g}$), again suggesting that the antimicrobial effect of AgNPs could exist for a long time.

3.5. Antibacterial properties of membrane surface

As illustrated by the disk tests with and without various membranes (Fig. 10), the original TFC and TFC-SH samples (b and c) had no significant effect on the growth of *E. coli*, while TFC-S-AgNPs (d and e) showed a clear LB surface with no evidence of bacterial growth.

An inhibition zone (0.5 mm) around the sample d was also observed. Similarly, many researchers [12,33] also found the appearance of inhibition zone around the substrate containing AgNPs due to the antibacterial function of AgNPs [34,35]. AgNPs could inhibit the growth of microbes through multiple pathways. Firstly, AgNPs can release Ag⁺ ions, which interact with disulfide

or thiol groups of enzymes or DNA and then disrupt metabolic processes, generate reactive oxygen species or interrupt replication of DNA [36–38]. All these processes could lead to the damage or even the death of bacterial cells. Secondly, AgNPs can also be attached to the surface of the cell membrane and disturb its proper function [35,39,40]. These studies also showed that nano-Ag could destabilize the outer membrane, collapse the plasma membrane potential and deplete the levels of intracellular ATP. Thirdly, AgNPs with smaller particle size (1–10 nm) are able to penetrate inside the bacteria and cause further damage by possibly interacting with sulfur- and phosphorus-containing compounds such as DNA [41]. However, due to the bigger size ($15.3 \pm 4.8 \text{ nm}$) and strong bonding between AgNPs and TFC-SH membrane surface, the third pathway may not be so significant in this study.

Notably, after 14-day silver release test, TFC-S-AgNPs membrane still showed good inhibition capability towards *E. coli*, which further confirmed the good stability of AgNPs on membrane surface.

Further disk experiment was carried out using real size membrane samples (OD = 3.7 cm, using for filtration test, Fig. 11). Similar to the observation in Fig. 10, after removing the membrane samples, growth of bacteria was observed under the pristine TFC membrane, while clear LB surface was observed under the TFC-S-AgNPs membrane. After checking the membrane surface contacted with bacteria, it is clear that spot-like *E. coli* colonies

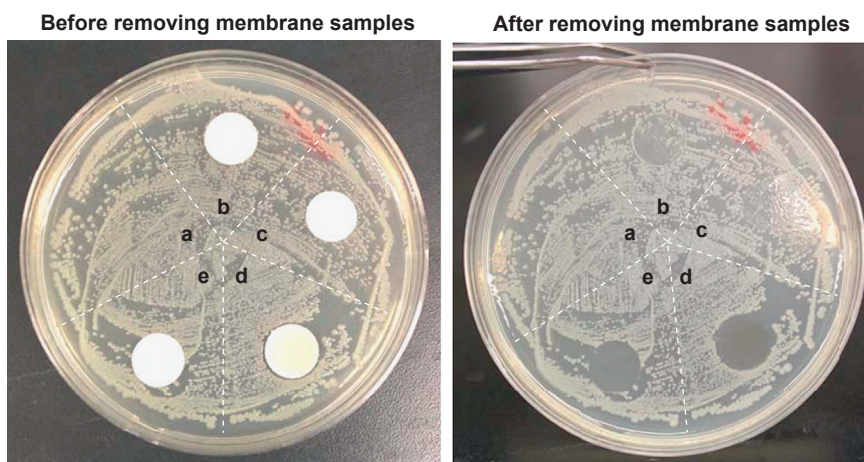


Fig. 10. Disk experiments for membranes of (a) control (without membrane sample); (b) TFC; (c) TFC-SH 24 h; (d) TFC-S-AgNPs and (e) TFC-S-AgNPs after 14-day batch release test. OD of membrane sample is 1.0 cm. Membranes were peeled from the nutrient agar after 24 h incubation.

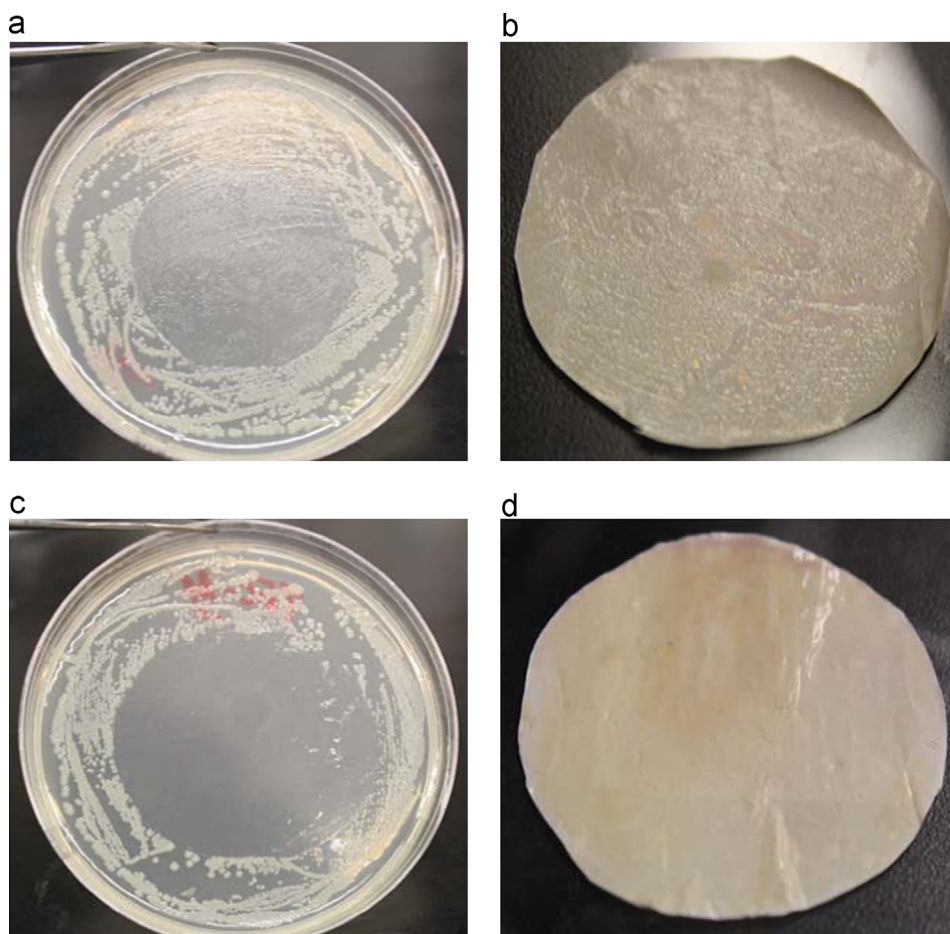


Fig. 11. Disk experiments: (a) area contacted with TFC; (b) TFC surface; (c) area contacted with TFC-S-AgNPs and (d) TFC-S-AgNPs surface. OD of membrane sample is 3.7 cm. Membranes were peeled from the nutrient agar after 24 h incubation.

were clearly observed on the TFC membrane surface, whereas the TFC-S-AgNPs membrane showed a clear and smooth surface. The easy attachment of *E. coli* on pristine TFC membrane surface suggests that PA TFC membrane surface is prone to biofouling and further demonstrates the necessity of surface anti-bacterial modification.

SEM images (Fig. 12) were obtained after removing membrane samples from the bacterial suspension. It was clear that rod-shaped *E. coli* bacteria with approximately 0.8 μm in diameter

and 2.0 μm in length attached on TFC membrane surface (indicated by white arrow in Fig. 12a) after 5 h contact. On the contrary, TFC-S-AgNPs membrane surface (Fig. 12b) was relatively clean and the attached cell was disrupted (magnified image in Fig. 12b) when compared with normal cell (magnified image in Fig. 12a). Gunawan et al. [5] also found the appearance of disrupted cells when they tested their Ag/MWNTs modified hollow fiber membrane using *E. coli*-containing feedwater, and attributed this phenomenon to the direct contact between the cells and AgNPs.

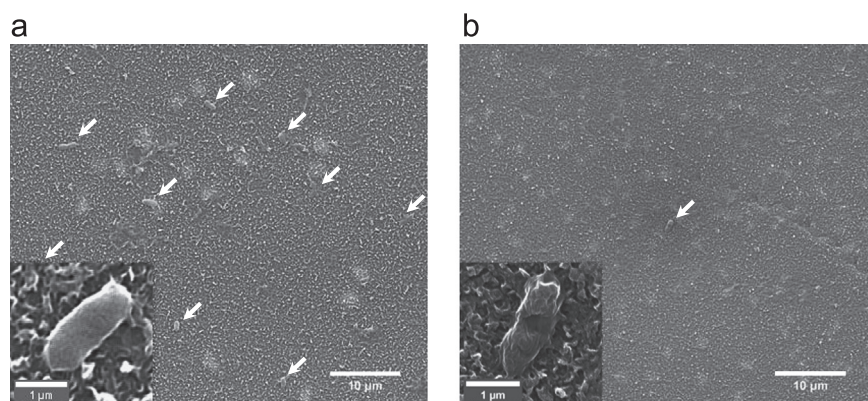


Fig. 12. SEM images of membrane surfaces submerged in an aqueous suspension of *E. coli* for 5 h: (a) TFC and (b) TFC-S-AgNPs.

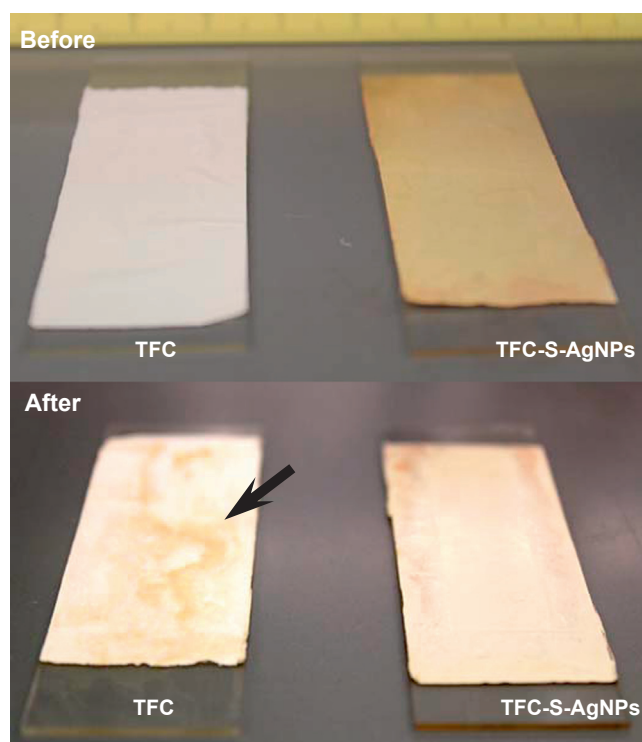


Fig. 13. Difference in *E. coli* biofilm growth on TFC membrane and TFC-S-AgNPs membrane in a drip flow biofilm reactor fed with a LB solution (5 g/L) at 1.0 ml/min for 7 days at 25 °C.

So this result suggested again that TFC-S-AgNPs membrane possessed excellent anti-bacterial properties.

Furthermore, after 7-day biofilm growth test (Fig. 13), biofilm formation was observed (indicated by black arrow) on the TFC membrane surface, while the TFC-S-AgNPs membrane surface was relatively clean and free of biofilm growth. All these results demonstrated that AgNPs-grafted membrane possessed excellent antibacterial function, which could reduce biofouling [42].

4. Conclusions

A covalent bonding method was developed to attach silver nanoparticles (AgNPs) onto the surface of polyamide (PA) thin-film composite (TFC) membrane with cysteamine as a bridging agent. The TFC-S-AgNPs membranes showed good stability of immobilized AgNPs and excellent antibacterial properties, while maintaining good water flux and salt rejection. The results

demonstrated that chemically immobilizing AgNPs onto membrane surface could be an effective approach to reduce membrane biofouling. Further research is needed to optimize the dose of AgNPs to be grafted onto the TFC membrane and determine its long-term performance under continuous filtration conditions.

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